Problema #3. Resistor Ping-Pong

(a)

$$F_{1,2} = q E_{1,2} \quad (1)$$

Para $\bar{E}$ uniforme $E = \frac{V}{d}$

$\bar{E}$ ~~debido~~ en una placa debido a la otra:

$$E_{1,2} = \frac{1}{2} E = \frac{1}{2}\frac{V}{d} \quad (2)$$

Luego, de Gauß

$$\oint \bar{E} \cdot d\bar{A} = \frac{Q}{\varepsilon_0}$$

$$\varepsilon_0 \frac{V}{d} (\pi R^2) = Q \quad (3)$$

(2) y (3) en (1)

$$\boxed{F_{1,2} = \frac{\varepsilon_0 \pi R^2}{2} \frac{V^2}{d^2}}$$

(b) De Gauß

$$\oint \vec{E} d\vec{A} = \frac{q}{\varepsilon_0}$$

$$\iff q = \varepsilon_0 E (\pi r^2)$$

$$\iff q = \varepsilon_0 \frac{V}{d} \pi r^2$$

Luego $q = \chi V$

$$\iff \chi = \frac{q}{V} = \frac{\varepsilon_0 \frac{\cancel{V}}{d} \pi r^2}{\cancel{V}}$$

$$\iff \boxed{\chi = \frac{\varepsilon_0 \pi r^2}{d}}$$

(c) ~~Dreieck~~ $F_e > F_g$

De (a)

$$F_e = \frac{\varepsilon_0 \pi r^2}{2} \frac{V^2}{d^2}$$

$$F_g = -mg$$

$$F_e = \underbrace{\frac{\varepsilon_0 \pi r^2}{d}}_{\chi} \frac{V^2}{2d} = \frac{\chi V^2}{2d}$$

$$F_e > F_g$$

$$\frac{\chi v^2}{2d} > mg$$

$$v^2 = \frac{2dmg}{\chi}$$

$$\boxed{v = \pm\sqrt{\frac{2dmg}{\chi}}}$$

(d)

$$\eta = (v_a / v_b)$$

$$v_s = (\alpha V^2 + \beta)^{1/2}$$

Coef. de restitución: indica la fracción de velocidad que se conserva después del choque. O sea, hay pérdida de velocidad → pérdida de Energía mecánica por colisión

colisión abajo

$$\Delta K = K_{\cancel{f}} - K_f$$

$$K_i = \frac{1}{2} m v_i^2$$

$$K_f = \frac{1}{2} m \eta^2 v_i^2 = \eta^2 K_i$$

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$$\Delta K = \eta^2 K_i - K_i$$

$$\Delta K = (\eta^2 - 1) K_i$$

pero $K_f = \eta^2 K_i \rightarrow K_i = \dfrac{K_f}{\eta^2}$

Entonces

$$\Delta K = (\eta^2 - 1) \frac{K_f}{\eta^2}$$

$$\Delta K = \left(1 - \frac{1}{\eta^2}\right) K_f.$$

Pero K_f es con v_s.

$$K_f = \frac{1}{2} m v_s^2.$$